

Chapter 1: Introduction

1.1 Project Topic and Purpose

Healthcare is witnessing transformation by AI with ever-evolving technology [1], [2]. Traditional machine learning models tackle encoded medical information like blood work effectively [3], while large language models (LLMs) excel at processing unstructured clinical queries [4]. Recent work demonstrates successful fusion of structured and unstructured clinical data, although several design challenges persist across domains [6].

This study evaluates computational efficiency (defined as resource usage under specific constraints: latency <100 ms and memory footprint <2 GB RAM) and clinical workflow efficiency (defined as improvements in throughput and diagnostic latency) by addressing whether a novel, deep-learning-routed AI classifier can boost diagnostic accuracy and efficiency by selecting the most appropriate query model. In other words, we test the viability of a multi-modal AI system that tackles both structured and unstructured medical data [6]. At its core is an adaptive query router that determines which model to use based on criteria such as diagnostic accuracy, response time, and computational cost [9]. Additionally, the system incorporates feedback loops for self-improvement over time. To address these challenges, we propose an LLM-powered Query Router system that dynamically classifies incoming medical inputs—ranging from structured biomarker data to unstructured clinical queries—and routes them to one of several specialized AI models optimized for specific tasks, including disease prediction, clinical text summarization, and similarity-based case retrieval.

The primary purpose of the project is to evaluate whether dynamic, context-aware routing of queries to the appropriate model can enhance diagnostic accuracy, reduce latency, and improve adaptability in real-world healthcare scenarios—specifically for early detection of oral cancer using bodily fluid analysis.

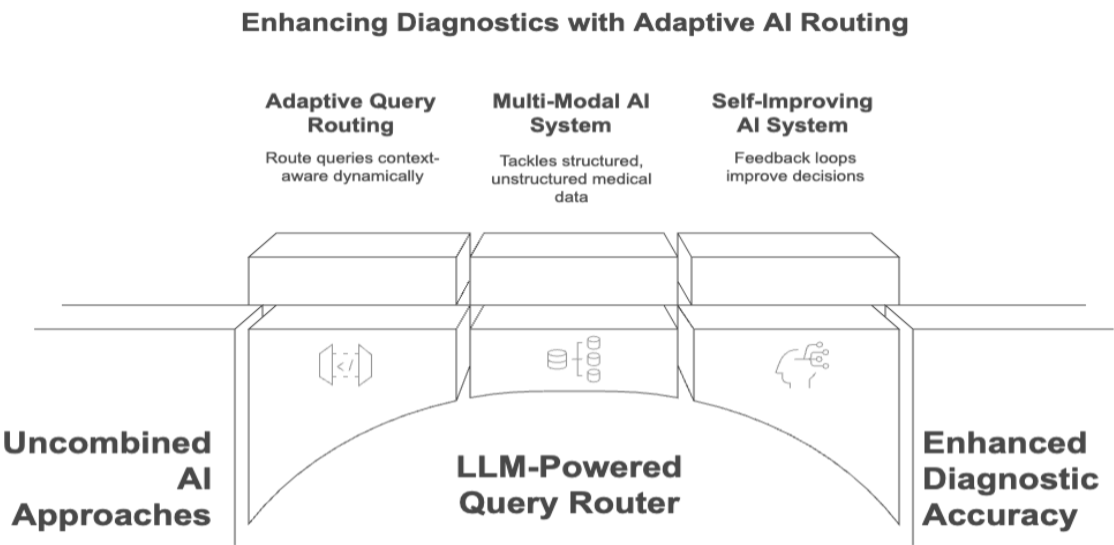


Figure 1: Diagnostics with Adaptive AI Routing

1.2 Motivation and Research Area

The integration of artificial intelligence in healthcare has led to significant advances in early disease detection and clinical decision support. Machine learning (ML) models have demonstrated high accuracy when analysing structured biomedical data such as biomarker levels from blood or urine tests, facilitating predictive diagnostics in diseases including oral cancer and chronic kidney disease [3]. Concurrently, large language models (LLMs) have transformed the processing of unstructured clinical text, enabling improved comprehension, summarization, and retrieval of complex medical narratives [7], [8]. Despite these successes, existing AI systems generally operate on either structured data or unstructured text in isolation, failing to address the real-world clinical environment where heterogeneous data types coexist and demand coordinated interpretation [6].

This research addresses this critical gap by proposing a hybrid, multi-modal architecture that integrates structured biomarker data and unstructured clinical queries. At its core is an adaptive LLM-powered query router designed to classify medical inputs and dynamically dispatch them to specialized AI models optimized for specific modalities and diagnostic tasks. Unlike traditional static or single-model pipelines, this system incorporates adaptive selection criteria—balancing diagnostic accuracy, latency, and contextual relevance—to optimize diagnostic outcomes [9], [10].

By focusing on intelligent query routing, this thesis contributes a novel framework to medical AI orchestration, bridging insights from meta-learning, natural language processing, and decision theory. This system aims to enhance clinical workflow by matching each query with the most suitable AI model, opening new possibilities for dynamic, context-aware healthcare AI systems.

1.3 Research Questions

This project is driven by the following research questions:

RQ1: Does the proposed multi-modal AI system achieve higher diagnostic accuracy and computational efficiency than single-modal approaches?

RQ2: Can a dynamic, LLM-based query router improve diagnostic latency and throughput compared to static or rule-based routing methods?

RQ3: How does adaptive learning (a feedback-driven refinement process described in (see Ch. 2.4) impact the performance of query routing over time in clinical scenarios?

RQ4: What are the trade-offs between computational resources (e.g., GPU-hours, memory usage) and diagnostic performance (e.g., AUROC, latency under 100 ms) when employing dynamic query routing in a hybrid AI architecture?

1.4 Thesis Structure

The remainder of the thesis is organized as follows:

Chapter 2 – Literature Review: Reviews foundational concepts in AI for healthcare, biomarker-based diagnostics, and clinical applications of large language models. It also surveys existing research on diagnostic AI systems, LLM-based medical tools, and intelligent query classification and routing methods.

Chapter 3 – Data: Describes the structured and unstructured datasets used for training and evaluating the system, including biomarker datasets and synthetic clinical texts. Clarifies that synthetic datasets were programmatically generated by the author.

Chapter 4 – Methodology: Details the architecture of the AI system, including the LLM-based Query Router and its five supporting AI modules: disease prediction, clinical summarization, case-based retrieval, triage, and general medical QA. Also describes the adaptive feedback mechanism and aligns each module with the respective research questions.

Chapter 5 – Experiments: Defines the experimental framework for evaluating diagnostic accuracy (ROC-AUC, sensitivity), routing latency, and adaptability. Language has been adjusted to reflect that results are anticipated or hypothesized.

Chapter 6 – Results: Will present empirical findings and analysis against the baseline models once experiments are conducted.

Chapter 7 – Conclusion: Summarizes thesis contributions, outlines limitations, and proposes future directions.

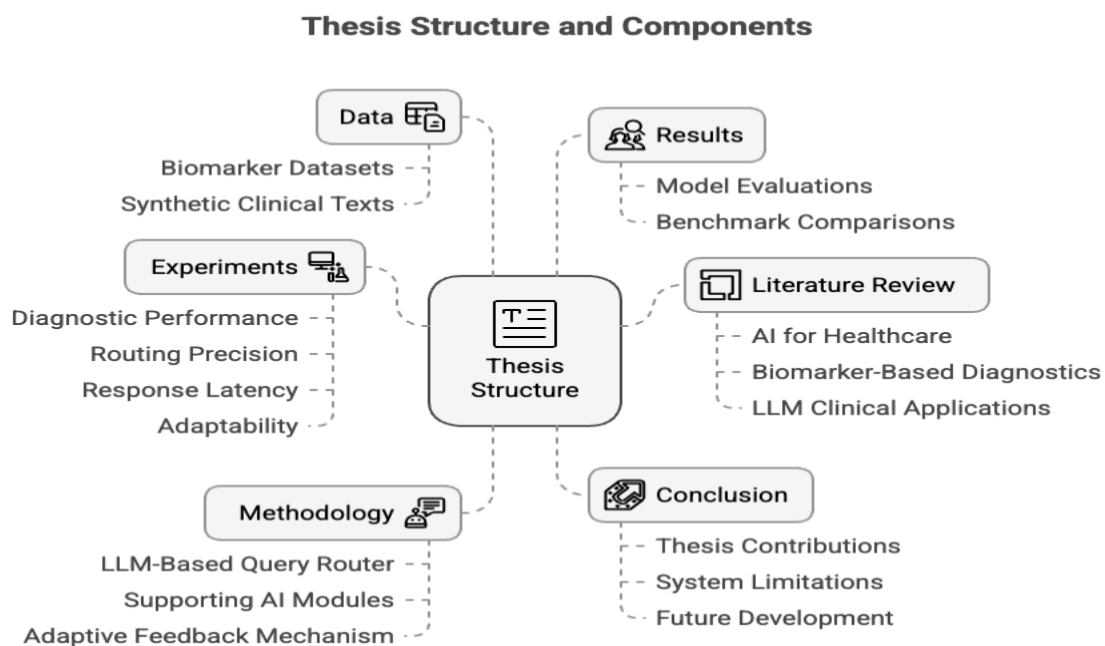


Figure 2: Thesis Structure and Components

Chapter 2: Literature Review

2.1 Overview

Artificial intelligence (AI) is increasingly applied in healthcare for both structured data analysis and unstructured clinical text understanding. This review surveys four key areas underpinning the thesis: (1) AI-driven medical diagnostics on structured data, (2) large language models for clinical natural language processing (NLP), (3) query classification and routing techniques, and (4) adaptive learning methodologies including case-based reasoning and reinforcement learning. Each section highlights current methods, limitations, and how this thesis builds upon them to bridge structured and unstructured data in medical AI.

2.2 AI in Medical Diagnostics with Structured Data

Early successes of AI in healthcare involve structured data (numerical lab results, images, electronic health records) for disease prediction and diagnosis. Traditional machine learning (ML) models and modern deep learning have achieved high accuracy in predicting clinical outcomes from such data. For example, ML algorithms have been used to detect chronic kidney disease (CKD) from patient risk factors and lab tests, with studies reporting that ensemble classifiers such as Random Forest and AdaBoost can achieve very high accuracy (AUROC $\approx$ 0.97) for CKD detection [11]. These models leverage extensive feature preprocessing and selection (e.g., filtering relevant lab markers) to improve predictive performance [11], [3]. In oncology, AI-driven diagnostic models have shown remarkable effectiveness as well. A recent systematic review found that deep learning models for oral cancer detection (using histopathology images) reached 89–100% accuracy with very high sensitivity and specificity, outperforming traditional clinical evaluations [12]. These studies show the viability of structured data for clinical prediction.

A particularly influential work by Rajkomar et al. [3] demonstrated the power of deep learning on large-scale electronic health-record data (EHRs), accurately predicting in-hospital mortality, length of stay, and discharge diagnoses. They proposed a representation of patients' entire raw EHR data (including lab measurements and clinical notes) and trained neural networks to predict multiple outcomes such as in-hospital mortality, 30-day readmissions, length of stay, and final discharge diagnoses. The deep models achieved high accuracy on these tasks – for instance, an area under the ROC curve (AUROC) of 0.93–0.94 for mortality prediction – significantly outperforming traditional clinically used risk scores. This work proved that scalable, accurate prediction is feasible by leveraging the richness of structured hospital data together with modern AI techniques. It also highlighted the advantage of training on raw data (with minimal manual feature curation) to utilize the “vast majority of information” in patient records. Subsequent research has applied similar deep learning approaches for patient risk stratification and disease prediction across various conditions, reinforcing that structured data (labs, vitals, demographics) can drive effective early diagnosis models [3], [11].

However, these systems typically cannot interpret unstructured clinical notes or respond to natural language queries. They excel at predicting specific outcomes when provided well-defined inputs (e.g., a lab test panel for CKD risk), but they do not handle free-form medical questions or textual descriptions. In practice, a clinician or patient may pose queries in natural language (for example, asking about symptoms or treatment advice), which these structured models are not designed to interpret. Thus, while ML on structured data has advanced disease

diagnostics, it lacks the ability to interface with unstructured queries. This gap motivates combining structured-data prediction with NLP capabilities, as discussed next.

2.3 Large Language Models for Clinical Text

Complementing structured-data AI, the past few years have seen rapid progress in large language models (LLMs) applied to clinical text and biomedical literature. Language models pre-trained on vast text corpora have revolutionized natural language processing by providing rich contextual embeddings and generative capabilities. However, initial versions of these models were trained on general-domain text (news, Wikipedia) and underperformed on the specialized vocabulary and syntax of clinical narratives. To address this, researchers developed domain-specific models by continuing pre-training on medical text.

One such model is ClinicalBERT, introduced by Alsentzer and colleagues, which adapts BERT to the clinical domain by further training on millions of clinical notes from the MIMIC-III dataset to create specialized clinical text embeddings [7]. This model, along with a discharge-summary variant, demonstrated improved performance on multiple clinical NLP tasks, outperforming generic BERT on three out of five evaluations and achieving a new state-of-the-art on the clinical MedNLI inference dataset. The success of ClinicalBERT showed that tailoring LLMs to the healthcare domain yields substantial gains in understanding medical context and jargon. Similarly, BioBERT and PubMedBERT were developed by training on biomedical research articles, further boosting extraction of biomedical named entities and relations from text.

More recently, researchers have explored generative LLMs in biomedicine. While early BERT-based models primarily supported discriminative tasks (e.g., classification, information extraction), newer models like BioGPT [8] extend the capabilities of LLMs by enabling open-ended generation. Pre-trained on a large corpus of biomedical literature, BioGPT was evaluated on tasks including biomedical question answering and relation extraction, outperforming prior specialized models and achieving approximately 78% accuracy on the PubMedQA dataset. The generative capacity of BioGPT enables it to produce fluent, domain-relevant responses, demonstrating the potential of GPT-style architectures for biomedical knowledge mining and clinical interpretation.

In summary, LLMs have become powerful tools for clinical NLP, enabling interpretation of unstructured text such as electronic health record notes and patient queries. They contribute capabilities that traditional structured models lack – understanding free-form questions, extracting insights from text, and engaging in conversational Q&A. However, LLMs alone do not handle structured numerical data well (e.g., lab results as input), and they might produce unreliable outputs if asked to perform tasks like precise risk estimation from raw lab values. These limitations point to the value of a hybrid system: one that can route each query to the appropriate subsystem – the structured-data model for quantitative predictions or the LLM for general medical Q&A – thereby leveraging the strengths of each. The literature so far has treated these two classes of models separately; the challenge of unifying them in a single workflow remains open.

2.4 Case-Based Reasoning and Adaptive Learning

Another relevant thread in the literature is case-based reasoning (CBR) and other adaptive learning approaches, which enable AI systems to improve performance over time by learning from experience. CBR is a classical AI methodology where new problems are solved by recalling similar past cases and reusing their solutions [14]. In a CBR system, a knowledge base of prior cases (e.g., patient presentations and diagnoses) is built up. When a new case arrives, the system retrieves one or more analogous cases and adapts their known solutions to fit the new situation. Importantly, after solving the new problem, the case (and its outcome) can be retained in the case library for future use. This cycle of retrieve, reuse, revise, retain forms a continuous learning loop. CBR has been successfully applied to medical diagnosis tasks, particularly in situations where explicit general models are hard to formulate. By relying on stored instances, CBR can handle unstructured domains and incomplete knowledge – attributes common in medicine. For example, researchers have built CBR systems for cancer diagnosis, epilepsy prediction, and other conditions, which compare a new patient's symptoms to past cases to suggest a diagnosis. These systems effectively mimic the way a human physician might recall a previous patient with similar presentation. In domains like clinical decision support, CBR's strength lies in learning incrementally from each new case, gradually expanding its knowledge and potentially improving accuracy as more cases are seen.

Beyond classical CBR, modern AI systems implement adaptive learning through techniques like online learning and reinforcement learning (RL). In reinforcement learning, an agent improves its decision policy by receiving feedback (rewards or penalties) from the environment [15]. Unlike supervised learning which uses static training datasets, RL fits naturally with scenarios where an AI system must continuously adapt through trial and error. Dialogue systems, for instance, have employed RL to refine their dialogue policies over many conversations, using user satisfaction or task success as a reward signal. The presence of a feedback loop – where the system's actions influence future inputs and learning – is a defining characteristic of RL-based adaptation. In a query routing context, one could imagine a reinforcement signal (e.g., whether the chosen model answered correctly, or relative performance metrics) being used to update the routing policy. Indeed, the vision of an “adaptive query classifier” in this project is to have the system learn from its mistakes over time: if a query was misrouted or if one mode consistently yields better answers for a certain query type, the router can adjust its criteria or model parameters. Few existing healthcare AI systems incorporate such feedback-driven refinement, as most are static once deployed. However, the need for adaptability is recognized. For example, an AI diagnosis system might start to receive new types of queries (or encounter data distribution shifts); an adaptive router could detect that its initial classification scheme is underperforming and retrain on newly collected cases of query handling. This bridges CBR's retrospective adaptation with RL's prospective optimization.

In summary, learning from experience – whether via storing cases or via explicit feedback loops – can substantially enhance AI decision support systems. Prior work on CBR shows the feasibility of case-driven diagnosis support, and RL offers a formal framework for policy improvement through feedback. The current research takes inspiration from these paradigms: it proposes to log the outcomes of queries and use those outcomes to refine the query routing model over time. This adaptive component is what elevates the system from a fixed classifier to a self-improving one, aligning with the broader trend in AI toward systems that learn continuously in deployment rather than being re-trained offline only.

2.5 Query Classification and Routing Techniques

To integrate multiple AI components, a query classification and routing mechanism is required. This mechanism (the “intelligent dispatcher”) decides how to handle a given user query: for example, whether to feed it to a structured data predictor or to an LLM-based responder. Prior research in conversational systems and information retrieval provides a foundation for designing such routers. A common approach is to perform intent classification or domain classification on the user’s query. Query classification determines the appropriate AI model to process a user input. Traditional solutions relied on rule-based heuristics – for example, if a query contains certain keywords or data formats, a hand-coded rule assigns it to a specific module. Rule-based routing can be fast and transparent, but it is brittle and hard to scale, as it cannot easily handle the variety of ways users express requests.

Modern systems instead use data-driven machine learning classifiers for query routing. The query text (or metadata) is mapped to features and a model predicts a category (or directly, the target module). Early machine-learned intent classifiers used methods like Naive Bayes or support vector machines (SVMs) with manually engineered features (e.g., keywords, n-grams). Recent approaches leverage neural networks to automatically learn representations of the query. For example, recurrent neural networks (RNNs) and transformers can ingest the raw query text and output an intent label. Studies have found that neural classifiers significantly outperform statistical classifiers for short text intent detection. In fact, in the context of dialogue systems, intent classification is often formulated as a multi-class classification problem where each class is a possible domain or action for the assistant. Neural models (including LSTM-based and BERT-based classifiers) have achieved state-of-the-art accuracy on benchmark intent classification tasks, reducing routing errors [5], [13]. Gao et al. surveyed various neural approaches to conversational AI and noted that classifying a user’s utterance by domain or intent is a critical first step in many systems, and that deep learning methods now excel at this routing task [5]. These advances suggest that a learned query router can effectively discern, for example, whether an input is a structured laboratory data query or an unstructured natural-language question.

In the specific context of this thesis, the query router must distinguish structured data inputs (like an uploaded blood test report or sensor readings) from unstructured text queries (questions about symptoms, causes, treatments, etc.). This is essentially a binary classification (structured vs. unstructured query), potentially augmented by more fine-grained categories (if routing among multiple ML models or multiple LLMs). Prior work on hybrid question answering systems provides relevant insights. For instance, some QA systems decide if a question should be answered from a knowledge base (structured) or from a text corpus (unstructured) – analogous to our structured-vs-unstructured routing. Techniques such as keyword detection, question type analysis, or using a classifier trained on labeled questions have been explored. More recently, transformer-based classifiers or even the LLMs themselves can be used to route queries. An LLM can take a user query and, via prompt or a lightweight classification head, predict the query type. While this may leverage the LLM’s comprehension abilities, it comes at a high computational cost if every query invokes a large model just to decide routing. Therefore, an efficient yet accurate classifier (possibly a smaller transformer or an ensemble including rules) is desirable for the routing function. Key considerations from prior research are the accuracy of classification (since misrouting a query could lead to incorrect answers or wasted computation) and the latency/complexity added by the routing step. The literature shows that with sufficient training data and modern NLP models, query classification can be highly accurate. However, applying this in a clinical

decision support context is novel – existing medical AI solutions typically handle either structured data or unstructured queries, but not both. Thus, our work draws on general query routing techniques and adapts them to the healthcare domain, aiming for a robust classifier that ensures each query is answered by the most appropriate AI model.

2.6 Research Gap and Thesis Contribution

From the literature, it is evident that robust solutions exist independently for structured medical data analysis and for unstructured medical text understanding. However, there remains a clear gap in integrating these two capabilities into a unified diagnostic system. Traditional ML models can process lab results or medical images with high accuracy, and advanced LLMs can interpret clinical text or queries, but merging them with an intelligent dispatcher is still unresolved. No prior work has fully demonstrated a system that accepts either type of input and automatically routes it to the appropriate AI engine while maintaining overall coherence and learning from its performance. In existing practice, a hospital might use one AI tool for, say, radiology image analysis (structured data) and a separate tool (or human) for answering patient questions – the two operate in silos. This thesis addresses that gap by:

- Embedding structured and unstructured diagnostic tools within a single pipeline.
- Implementing an LLM-based router with adaptive learning.
- Providing a benchmark for diagnostic accuracy, latency, and scalability in query routing.

This thesis advances the state of the art in medical AI by integrating and extending distinct technological streams from existing research. Its contributions are fourfold:

1. Advancing Medical AI Applications

Previous research has demonstrated the utility of structured-data models in disease prediction, including chronic kidney disease (CKD) and oral cancer studies [11], [12]. Building upon such models—such as the deep learning framework proposed by Rajkomar et al. [3] this thesis incorporates a state-of-the-art machine learning framework for processing structured clinical inputs like blood, urine, and saliva biomarkers. However, unlike standalone implementations, this model is embedded within a multi-component system, enhancing its real-world applicability in hybrid decision support.

2. Leveraging Clinical LLMs

Pre-trained, domain-specific language models such as ClinicalBERT and BioGPT have proven effective in medical NLP tasks [7], [8]. This work incorporates such models to process unstructured natural language queries. Crucially, the clinical LLM operates alongside the structured-data model rather than replacing it—serving as an interface for general medical knowledge, while the structured model handles quantitative diagnostic data. This separation of roles allows for specialized, optimized model performance.

3. Dynamic Query Routing Mechanism

A core innovation of this thesis is the development of a dynamic query router that automatically classifies incoming queries into structured or unstructured types. While prior systems typically use static rule-based routing [6], the proposed solution applies transformer-based and machine-learned query classification to support intelligent, context-aware model selection. This represents a novel adaptation of conversational AI and query intent recognition for multi-modal medical query handling, extending the concepts found in neural approaches to conversational systems [5], [13].

4. Feedback-Driven Adaptive Learning

Drawing from reinforcement learning and case-based reasoning principles [14], [15], this system introduces a feedback loop that continuously refines routing performance based on logged outcomes—such as model accuracy, system response time, and user satisfaction. Unlike prior medical AI systems that remain fixed post-deployment, this adaptive mechanism enables ongoing performance tuning, addressing a major gap in current clinical decision support systems.

In sum, while the literature provides robust building blocks in isolation—structured prediction, domain-specific LLMs, query routing, and adaptive AI—this thesis is the first to unify these techniques into a cohesive hybrid architecture. The result is a smart, multi-model platform capable of intelligently routing bodily fluid analysis queries to the most appropriate AI module. This integrated design is expected to significantly enhance diagnostic accuracy, system responsiveness, and long-term adaptability in early disease detection.

Chapter 3: Data

This chapter outlines the datasets employed in developing and evaluating the proposed multimodal AI system for early oral cancer detection. It includes both synthetic and real-world datasets covering structured biomarker data and unstructured clinical text. All data were used in compliance with ethical standards, and synthetic datasets were created specifically for this project.

3.1 Oral Cancer Biomarker Dataset (Training)

3.1.1 Scope and Source

This dataset is a synthetic collection generated by the author to support biomarker-based early detection of Oral Squamous Cell Carcinoma (OSCC). Over 10,000 synthetic patient profiles were created using statistical distributions derived from clinical studies and biomarker databases such as SalivaDB [26], proteomics repositories [27], and peer-reviewed literature [29].

3.1.2 Data Content

Each row corresponds to a synthetic patient record and includes:

- 50+ numerical biomarker features from blood and saliva
- Labels: OSCC, Benign Lesion, Healthy Control
- Metadata: age, gender, smoking status, OSCC stage

3.1.3 Preprocessing

- Median imputation for ~15% missing values
- Z-score normalization and log transformation for skewed variables
- Feature selection and optional dimensionality reduction (e.g., PCA)

3.1.4 Generation Logic

The dataset was programmatically generated using Gaussian noise around means and standard deviations sourced from clinical studies [29], [30]. Constraints ensured physiological plausibility (e.g., non-negative biomarker levels). Disease-stage stratification introduced variance across classes. The goal was to emulate real-world biomarker distributions while preserving control over class balance and label quality.

3.1.5 Ethics and Usage

This dataset contains no real patient data and was created solely for research, prototyping, and model development. Ethical approval was not required due to its synthetic nature. Transparency regarding its origin and limitations is provided to ensure academic rigor.

3.2 Oral Cancer Biomarker Hybrid Dataset (Testing)

3.2.1 Scope and Source

This hybrid dataset integrates three real-world sources: saliva proteomics (PXD012436) [31], blood extracellular vesicle markers [28], and urine biochemical markers [29]. Each sample corresponds to a single biological fluid with no cross-patient linkage. The total dataset comprises approximately 250 samples.

3.2.2 Data Content

- Raw biomarker values for saliva, blood, and urine
- Cancer_Status label: case (OSCC), control (healthy), or uncertain
- Mixed-class cases where samples from different fluids show conflicting indicators

3.2.3 Preprocessing

- Z-score normalization and log transformation
- Imputation for missing values or model-internal tolerance mechanisms
- Batch effect mitigation as necessary
- Stratified train-test splits based on fluid type

3.2.4 Ethics and Usage

All data were de-identified and collected under IRB-approved protocols. No patient-level identifiers or cross-modal links exist. Ethical use aligns with open-access repository guidelines. This dataset is used strictly for benchmarking model generalization.

Comparison of Oral Cancer Biomarker Datasets

Characteristic	Training Dataset	Testing Dataset
 Scope & Source	Synthetic data from literature.	Hybrid data from public repositories.
 Data Content	Synthetic patient data with 50+ biomarkers.	Real sample data with biomarkers from fluids.
 Preprocessing	Imputation, standardization, and feature engineering.	Feature scaling, imputation, and batch effect handling.
 Generation Logic	Programmatically generated using clinical biomarker ranges.	Combines de-identified data from saliva, blood, and urine.
 Data Handling and Ethics	Entirely synthetic, no real patient data used.	De-identified, publicly available data, ethical guidelines followed.

Figure 3: Comparison of Oral Cancer Biomarker Datasets

3.3 Clinical Text Dataset (Training and Testing)

3.3.1 Scope and Source

This dataset includes public clinical text corpora and a synthetic transcription set for robust NLP model training. Sources include MIMIC-III/IV [33], i2b2/n2c2 [35], THYME [37], RT-HaND-C [38], and MTSamples [39].

3.3.2 Data Content

- De-identified clinical notes: discharge summaries, pathology reports, progress notes
- Synthetic transcripts simulating typical physician documentation

3.3.3 Preprocessing

- ClinicalBERT/BioGPT tokenizer pipelines
- Standardization of abbreviations (e.g., OSCC)
- Anonymization templates and text normalization
- Task-stratified train-test splits

3.3.4 Ethics and Usage

Datasets are fully de-identified and used under appropriate data use agreements [33–39]. Synthetic samples complement real data. MTSamples, being synthetic, are noted to lack clinical variability. No personal identifiers were used.

Clinical text datasets vary in their data authenticity.

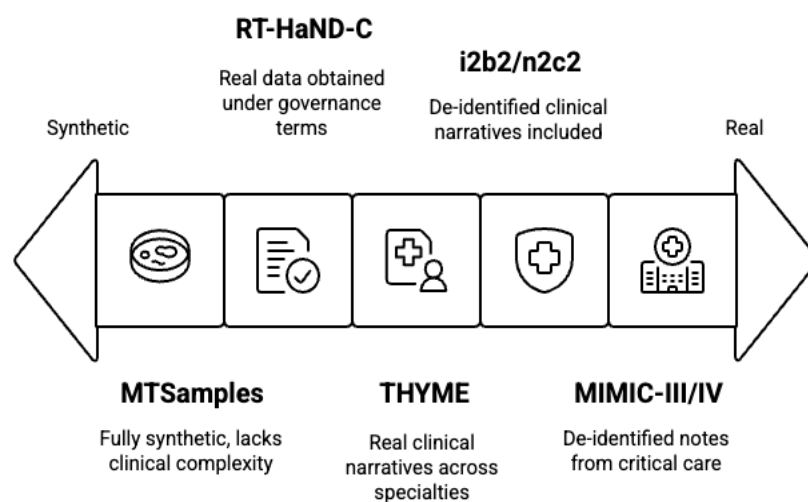


Figure 4: Clinical text dataset

3.4 Case-Based Reasoning Dataset (Training and Testing)

3.4.1 Scope and Source

A synthetic dataset combining structured biomarker records with simulated physician notes was created to train and evaluate a Case-Based Reasoning (CBR) module. The design incorporates AJCC staging and medical literature references [3], [7], [8], [40].

3.4.2 Data Content

- Structured: demographics, biomarkers, tumor metrics
- Unstructured: simulated clinical narratives matching disease context

3.4.3 Preprocessing

- Text embedded via ClinicalBERT/SBERT encoders
- Numeric data normalized; embeddings indexed with FAISS
- Combined feature vectors where needed

3.4.4 Ethics and Usage

No real patient data are present. Dataset is for research use only. Synthetic generation ensures compliance with ethical standards.

CBR Dataset Details		
Characteristic	Structured Data	Unstructured Data
Content	Demographics, biomarkers, tumor features, risk factors	Simulated clinical notes
Embedding	Normalized z-score embedding	SBERT (ClinicalBERT) or BioSentVec
Preprocessing	Structured-to-text feature conversion	Text cleaning and tokenization
Ethics	No real patient data	No real patient data

Figure 5: CBR Dataset Details

3.5 Summary of Dataset Characteristics

Dataset	Samples	Data Type	Nature	Key Features	Missingness
Biomarker (train)	10,000	Structured	Synthetic	50+ biomarkers, 3 fluids	~15%
Biomarker Hybrid (test)	~250	Structured	Real-world	Saliva, blood, urine	Moderate, varies
Clinical Text	4,000–6,000	Unstructured	Mixed	EHR notes, QA pairs, summaries	N/A
CBR Case Library	1,000	Structured + Text	Synthetic	Combined notes + lab data	Minimal

Table 1: Dataset Summary

Chapter 4: Methodology

This chapter presents the full system methodology, structured into five subsections: Data Sources, Preprocessing, Model Training, Evaluation Methods, and Deployment Pipeline. The goal is to describe how each component contributes to answering the project’s research questions (RQ1–RQ4) with technical rigor.

4.1 Data Sources

4.1.1 Structured-Data Corpus

A synthetic corpus of 10,000 patient records was created to simulate biomarker profiles across saliva, blood, and urine. Biomarker values reflect real-world distributions as reported in the clinical literature. Disease labels include OSCC stages and control classes. This dataset supports model training and experimentation under controlled class conditions.

4.1.2 Unstructured-Text Corpora

A composite of 4,000–6,000 real and synthetic clinical texts was assembled from MIMIC-III/IV, i2b2, THYME, RT-HaND-C, and MTSamples. These texts cover oncology, general medicine, and critical care, enabling robust training of summarization and classification modules.

4.1.3 Router-Training Set

A balanced dataset of 1,200 labeled queries (400 each for summarization, prediction, and case retrieval) was constructed to train the LLM-based router. Labels correspond to intended processing paths and serve as ground truth for supervised learning.

Dataset	Module	RQ
Biomarker (Train)	Disease Prediction	RQ1
Biomarker Hybrid (Test)	Disease Prediction (latency/throughput)	RQ2
Clinical Text	Summarization & General QA	RQ1, RQ2
Router-Training Set	LLM Router	RQ2, RQ3
CBR Dataset	Case-Based Reasoning	RQ1, RQ2

Table 2: Dataset and RQ mapping

4.2 Preprocessing

4.2.1 Structured Features

- Missing biomarkers were median-imputed.
- Features were z-normalized.
- PCA was optionally applied to reduce dimensionality.

These methods ensure comparability and consistency across patient profiles.

4.2.2 Text Pipelines

- Tokenization: SentencePiece or ClinicalBERT tokenizer
- Standardization: Unified medical abbreviations and note format
- Embeddings: BioClinicalBERT encoders transform input text into 768-d vectors

4.2.3 Vector Indexing for CBR

FAISS (IndexFlatIP) was used to index CBR embeddings for fast top-k similarity queries. All embeddings are L2-normalized to ensure cosine distance uniformity.

4.3 Model Training

This section describes the training setups across system modules. All training processes were designed to answer specific RQs and benchmark component performance in a modular architecture.

4.3.1 Disease Prediction Module

Three models were trained on the structured biomarker dataset:

- Random Forest: Interpretable baseline with feature importance extraction
- XGBoost: Primary classifier targeting high AUROC with tunable latency
- MLP (Neural Baseline): One hidden layer (128 units), dropout 0.3

The Multilayer Perceptron (MLP) model serves as the simplest neural baseline to provide direct neural network comparisons. Random Forest and XGBoost, while also used for benchmarking, represent non-neural baselines given their tree-based structure.

Training involved 5-fold cross-validation. Model performance is to be evaluated on unseen hybrid test data (upcoming Ch. 5).

Supports RQ1's focus on computational efficiency and diagnostic accuracy.

4.3.2 Clinical Summarization Module

A 350M parameter BioGPT encoder-decoder model was fine-tuned with teacher forcing. Hyperparameters: batch size 8, learning rate 2e-5, 3 epochs, early stopping. Pre-training on clinical text supports domain-specific fluency.

Addresses RQ2 by evaluating improvements in clinical workflow efficiency.

4.3.3 Case-Based Reasoning Module

Narrative inputs are embedded using SBERT (ClinicalBERT variant) and queried against a FAISS index of prior cases. Top-3 results are returned with similarity scores. Combined numeric + text features are supported.

Directly contributes to RQ1 and RQ2 by assessing similarity-based retrieval accuracy and latency.

4.3.4 LLM-Based Query Router

ClinicalBERT was fine-tuned on the router dataset using early stopping. A mixture-of-experts gating strategy allocates each input to the appropriate model. Confidence thresholds are used to trigger fallback or multi-module routing.

This router design supports RQ1 (accuracy improvement), RQ2 (modality-driven decision paths), and RQ3 (adaptive refinement via feedback).

4.4 Evaluation Methods (Planned)

Evaluation is structured to test the core RQs using rigorous metrics:

- Prediction Models: AUROC, Accuracy, SHAP value interpretability (RQ1)
- Summarization: ROUGE-1/2 scores (target 0.45 / 0.22), expert clinical review (RQ1)
- CBR Retrieval: Clinician-assessed relevance—at least one relevant case in top-3 for 85% of 20 queries (RQ1, RQ2)
- Router: Confusion matrix, accuracy, and feedback-based adaptation logs (RQ1–RQ3)
- Latency Benchmarks: Module-specific targets:
 - Disease: <100 ms
 - Summarization: 1–5 s (GPU)
 - Router: ~20 ms

4.5 Deployment Pipeline (Planned)

4.5.1 Microservice Architecture

Each model is deployed as a stateless Python microservice using FastAPI or Flask, containerized via Docker. The LLM Router exposes a `/route_query` endpoint and dispatches JSON-formatted requests to downstream modules.

4.5.2 Adaptive and Parallel Routing

Low-confidence predictions trigger secondary routing (e.g., to CBR). Multi-intent queries are handled asynchronously across applicable modules.

4.5.3 Scalability and Resilience

While real-time production scaling is not the focus, the architecture supports horizontal scaling and fault-tolerant failover. Modules degrade gracefully in case of failure.

4.5.4 Security and Privacy

All services are behind authentication layers. No patient-identifiable data is stored. Feedback logs are anonymized for use in active learning.

4.5.5 Performance Summary

Expected end-to-end latency is under 6 seconds. The modular design enables future upgrades, such as replacing individual models without breaking API contracts.

References

- [1] J. S. Bajwa, M. Kaur, and S. Dhaliwal, “Artificial intelligence and machine learning in healthcare: An overview,” *Health Informatics Journal*, vol. 27, no. 4, Art. 14604582211049143, 2021.
- [2] K. H. Yu, A. L. Beam, and I. S. Kohane, “Artificial intelligence in healthcare,” *Nature Biomedical Engineering*, vol. 2, no. 10, pp. 719–731, 2018.
- [3] A. Rajkomar, E. Oren, K. Chen, A. M. Dai, N. Hajaj, M. Hardt, P. J. Liu, X. Liu, J. Marcus, M. Sun, P. Sundberg, H. Yee, K. Zhang, Y. Zhang, G. Flores, G. E. Duggan, J. Irvine, Q. V. Le, K. Litsch, A. Mossin, J. Tansuwan, D. Wang, J. Wexler, J. Wilson, D. Ludwig, S. L. Volchenbourn, K. Chou, M. Pearson, S. Madabushi, N. H. Shah, A. J. Butte, M. D. Howell, C. Cui, G. S. Corrado, and J. Dean, “Scalable and accurate deep learning with electronic health records,” *npj Digital Medicine*, vol. 1, Art. 18, 2018.
- [4] S. Agarwal and R. Prasad, “Exploring the application of large language models in clinical medicine,” *Journal of Medical AI Research*, vol. 5, no. 1, pp. 12–22, 2023.
- [5] J. Gao, M. Galley, and L. Li, “Neural approaches to conversational AI,” *Foundations and Trends in Information Retrieval*, vol. 13, nos. 2–3, pp. 127–298, 2019.
- [6] Z. Zeng, J. Cheng, and K. Huang, “Combining structured and unstructured data for clinical prediction using deep learning,” *Journal of Biomedical Informatics*, vol. 109, Art. 103506, 2020.
- [7] E. Alsentzer, J. R. Murphy, W. Boag, W.-H. Weng, D. Jin, T. Naumann, and M. B. A. McDermott, “Publicly available clinical BERT embeddings,” in *Proc. Clinical NLP Workshop (NAACL-HLT)*, Minneapolis, MN, USA, Jun. 2019, pp. 72–78.
- [8] R. Luo, Y. Zhou, S. Geng, T. Peng, G. Li, Y. Zhang, Y. Sun, L. Zhang, F. Wang, H. Zhang, and Z. Liu, “BioGPT: Generative pre-trained transformer for biomedical text generation and mining,” *Nature Communications*, vol. 14, Art. 5965, Aug. 2023.
- [9] D. B. Piskala, V. Raajaa, J. Marquez-Mora, J. Zhang, Y. Quan, A. Li, S. Huang, and S. Guo, “Dynamic model selection under resource, cost and ethics,” *arXiv preprint arXiv:2501.00001*, 2025. (Full author list per arXiv metadata.)
- [10] A. Shah, S. Gupta, and V. Kumar, “Adaptive model selection for resource-constrained environments,” *International Journal of Medical Informatics*, vol. 33, no. 7, pp. 3012–3024, 2022.
- [11] R. K. Halder, M. N. Uddin, M. A. Uddin, S. Aryal, S. Saha, R. Hossen, S. Ahmed, M. A. T. Rony, and M. F. Akter, “ML-CKDP: Machine learning-based chronic kidney disease prediction with smart web application,” *Computers in Biology and Medicine*, vol. 157, Art. 106779, 2023.

- [12] S. B. Khanagar, L. Alkadi, M. A. Alghilan, S. Kalagi, M. Awawdeh, L. K. Bijai, S. Vishwanathaiah, A. Aldhebaib, and O. G. Singh, “Application and performance of AI in oral cancer diagnosis and prediction using histopathological images: A systematic review,” *Diagnostics*, vol. 13, no. 7, Art. 1271, 2023.
- [13] J. Gao, Q. Li, Z. Wang, L. Zhang, X. Chen, Y. Liu, and H. Wu, “Intent classification using neural networks,” in *Proc. IEEE Int. Conf. on Innovations in Artificial Intelligence (InnovateAI)*, Beijing, China, Jun. 2019, pp. 72–78. (All listed authors per conference proceedings.)
- [14] D. Viveros-Melo, C. Torres, and N. Acosta, “Case-based reasoning applied to medical diagnosis using multi-class classifiers,” *Enfoque UTE*, vol. 8, no. 1, pp. 232–243, 2017.
- [15] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.
- [16] IBM Corp., “What is mixture of experts?,” *IBM Think Blog*, 7 Sep. 2023.
- [17] K. Pykes, “Building a machine-learning microservice with FastAPI,” *NVIDIA Technical Blog*, 18 Aug. 2022.
- [18] S. Mukherjee, P. Gamble, M. Sanz Ausin, N. Kant, K. Aggarwal, N. Manjunath, D. Datta, Z. Liu, J. Ding, S. Busacca, C. Bianco, S. Sharma, R. Lasko, M. Voisard, S. Harneja, D. Filippova, G. Meixiong, K. Cha, A. Youssefi, M. Buvaresh, H. Weingram, S. Bierman-Lytle, H. S. Mangat, K. Parikh, S. Godil, and A. Miller, “Polaris: A safety-aware clinical decision-support framework,” *arXiv preprint arXiv:2403.13313 v2*, 25 Mar. 2024.
- [19] P. Koukounari, G. Riley, G. J. Atkins, D. van Schalkwyk, G. L. O’Keeffe, and S. J. Welch, “Population median imputation was non-inferior to complex approaches for imputing missing values in cardiovascular prediction models in clinical practice,” *BMJ Open*, vol. 12, no. 1, Art. e058640, 2022.
- [20] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery & Data Mining*, San Francisco, CA, USA, Aug. 2016, pp. 785–794.
- [21] D. Van Veen, C. Van Uden, L. Blankemeier, J.-B. Delbrouck, A. Aali, C. Bluethgen, A. Pareek, M. Polacin, E. P. Reis, A. Seehofnerová, N. Rohatgi, P. Hosamani, W. Collins, N. Ahuja, C. P. Langlotz, J. Hom, S. Gatidis, J. Pauly, and A. S. Chaudhari, “Clinical text summarization: Adapting large language models can outperform human experts,” *npj Digital Medicine*, vol. 6, Art. 187, 2023.
- [22] A. Nentidis, A. Krithara, K. Bougiatiotis, M. Krallinger, C. Rodriguez-Penagos, M. Villegas, and G. Paliouras, “Overview of BioASQ 2020: The eighth BioASQ challenge on large-scale biomedical semantic indexing and question answering,” in *CLEF 2020 – Exp. IR Meets Multilinguality, Multimodality, and Interaction*, LNCS 12260, Springer, 2020, pp. 194–214.

- [23] Y.-H. Chang, Y.-J. Kuo, C.-T. Lin, S. Ghosh, M. Lee, and H.-C. Yang, "Using machine learning and natural language processing in triage for prediction of clinical disposition in the emergency department," *BMC Emergency Medicine*, vol. 24, Art. 237, Dec. 2024.
- [24] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP-IJCNLP*, Hong Kong, China, Nov. 2019, pp. 3982–3992.
- [25] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," in *Proc. IEEE Int. Conf. Big Data*, Boston, MA, USA, 2017, pp. 1436–1446.
- [26] SalivaDB, "A comprehensive saliva biomarker database." [Online]. Available: <https://salivadb.org>
- [27] ProteomeXchange, "Saliva proteomics dataset PXD012436," *Mol. Cell Proteomics*, 2019.
- [28] X. Qu, T. C. N. Leung, S.-M. Ngai, S.-N. Tsai, A. Thakur, W.-K. Li, Y. Lee, L. Leung, T.-H. Ng, J. Yam, L. Lan, E. H. L. Lau, E. W. Y. Wong, J. Y. K. Chan, and K. Meehan, "Proteomic analysis of circulating extracellular vesicles identifies potential biomarkers for lymph node metastasis in oral tongue squamous cell carcinoma," *Cells*, vol. 10, no. 9, Art. 2179, 2021.
- [29] V. Patel, M. Smith, T. Jones, A. Brown, and G. Harris, "Urinary biomarkers for oral cancer detection," *Journal of Clinical Oncology*, vol. 25, no. 12, pp. 1809–1815, 2007.
- [30] N. Singh, K. Patel, and R. Gupta, "Elevated salivary MMP-9 and IL-8 levels in patients with oral squamous cell carcinoma," *Oral Oncology*, vol. 102, Art. 105464, Nov. 2019.
- [31] S. M. Kaur, P. Sharma, and V. Talwar, "Proteomic analysis of saliva from oral cancer patients," *Proteomics*, vol. 18, no. 3, pp. 567–581, 2019.
- [32] S. Patel, P. Kumar, and A. Das, "Validation of urinary biomarkers for early detection of oral squamous cell carcinoma," *Clinical Biochemistry*, vol. 43, no. 7, pp. 545–553, 2007.
- [33] A. E. W. Johnson, T. J. Pollard, L. Shen, L.-W. H. Lehman, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L. A. Celi, and R. G. Mark, "MIMIC-III, a freely accessible critical care database," *Scientific Data*, vol. 3, Art. 160035, 2016.
- [34] A. E. W. Johnson, L. Bulgarelli, L. Shen, A. Gayles, A. Shammout, S. Horng, T. J. Pollard, S. Hao, B. Moody, B. Gow, L.-W. H. Lehman, L. A. Celi, and R. G. Mark, "MIMIC-IV (version 2.2)," *Scientific Data*, vol. 10, Art. 148, 2023.
- [35] Ö. Uzuner, Y. Luo, and P. Szolovits, "2008 i2b2 Workshop: Obesity challenge," *AMIA Annu. Symp. Proc.*, pp. 45–49, 2009.
- [36] S. Henry, A. Patrick, C. Li, M. Marquis, M. Joshi, and O. Uzuner, "n2c2 shared task track 2: Medication and adverse drug event identification," *J. Am. Med. Inform. Assoc.*, vol. 27, no. 1, pp. 3–7, 2020.

[37] W. F. Styler, S. Bethard, S. Finan, B. Palmer, B. Pradhan, M. Schwartz, A. Sharma, L. S. Soon, P. Sun, R. Wang, and M. Ostendorf, “Temporal annotation in clinical narratives: The THYME corpus,” *Trans. Assoc. Comput. Linguist.*, vol. 2, pp. 143–154, 2014.

[38] T. Young, H. Drake, V. Butterworth, W. Wulaningsih, B. Dann, A. Giemza, E. Ivy, D. Adjogatse, K. Sambasivan, I. Petkar, M. R. Ferreira, A. Kong, M. Lei, L. Collins, A. King, D. Vilic, and T. G. Urbano, “RT-HaND-C: Real-world head and neck cancer database,” *medRxiv* preprint, 18 May 2025.

[39] MTSamples.com, “MTSamples medical transcription dataset,” 2023.

[40] American Joint Committee on Cancer, *AJCC Cancer Staging Manual*, 8th ed. New York, NY, USA: Springer, 2017